

H O B B S S

NEW DISCOVERY

FOR LONGITUDE



ROYAL OBSERVATORY

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*** START OF THE PROJECT GUTENBERG EBOOK A NEW
DISCOVERY FOR FINDING THE LONGITUDE ***

A
New DISCOVERY

For Finding the
LONGITUDE.

Humbly Submitted to the

Approbation of the Right Honourable the Lords Spiritual and Temporal,
and the Rest of the Honourable Persons, appointed by the late *Act of
Parliament*, for Hearing and Determining *Proposals* relating to the
said LONGITUDE.

By William Hobbs, *Philo Mathem.*



LONDON:

Printed for the Author, and are to be Sold by *Fard. Burleigh* in *Amen-
Corner*.

Price Six-pence.

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A New Discovery for Finding the Longitude.

For Finding the Longitude, it is to be noted, That if the true Hour, and Minute of the Day, at the Place of our first Departure, and where we are Arrived, can be both obtained, the Difference of Longitude may be found, as truly as the Difference of Latitude.

For which purpose the *Spring-Movement* hereunto subjoined, is most humbly proposed to effect the same, even to the Tenth of a Minute of Time. So that should the Error of a Minute happen to be found in Practice, it must proceed from the Influence or Vicissitude of the Air, and not from any Defect in the Rules by which the said Movement is composed, as will plainly appear by the Explication thereof, in the following Lines; and first of

The Description of the said Movement.

This Movement may be either of the Form and Bigness of that in a Common Clock; or else of the Smallness of a Pocket Watch, as may be thought convenient in Practice. But the larger it is, the better the Divisions in the *Horologe* will appear. And as for the Numbers, they may be at pleasure: Only this must be observ'd, That the swiftest Index must make just One Revolution, whilst the slow one goes but One of the Hundred Divisions, (as is common in Minute Watches, wherein 60 Minutes goes to an Hour) but the rest of the Numbers may be at pleasure, as aforesaid. And if the Slowest moves round in 3, 4, or 5 Days; and the Movement requires to be wound up once in 26 or 27 Hours, it will be sufficient: For the more Vigor the Motion is calculated for, the less the Air will affect it.

How to set it first in Motion.

When it is finished, let it go for a Month or two; after which put the two *Indexes* right one with the other, and both of them pointing just to 100. Then the Spring being wound up, keep it from Going (with the Key) till the Shadow of the Sun is exactly come to a *Meridian* Line, which must be rightly prepared for that purpose. Then let it go till the Shadow comes just to the same Line the next Day, or rather 40, 50, or 100 Days after, (the longer the Better.) At which time enter down the *Centesms* that the swiftest *Index* points to, as also the *Integers* and *Tenths* of the slowest, (but put the last first in your Numbers) by this you will be furnished with two general Numbers, viz. the said Numbers pointed to, and the Time (in Hours) spent therein, to find the Hour of the Day to the Tenth of a Minute, at any time; for that Place where it was so set going, tho' you remove it afterwards to any Distance whatsoever. Which two Numbers ought to be entered in a Book; to be used whensoever you would find the Hour of the Day.

And it is to be noted, That if the Sun be not then in its Mean Motion, you must Add to, or Subtract from what the *Indexes* do Give, according to the Inequality thereof: But if you try it for one whole Year, there will be no need of either. And that what is said may be the better understood, I shall give an Example.

How to find the Hour and Minute of the Day at London, by the said Movement.

Suppose it should be made for the Slowest to Revolve in about five Days, and after it has gone just two Days, the Slowest *Index* should then point between 39 and 40, and the Swiftest to 21.5. Then both these Numbers (as set in Order) will make 3921.5. And if the Sun be in its Mean Motion (if not, you must Add or Subtract, as aforesaid) then the said Numbers 3921.5 must always be the First; the said two Days, or 48 Hours, the Second; and the two Numbers pointed to by the *Indexes*, at the time for which you would find the Hour and Minute of the Day (as suppose the slow one should point to 87, and the swift one to 65.2, both making 8765.2) must be the Third; by which the Fourth will be obtained as followeth: Which Fourth Number being Divided by 12, the Remainder will be the Hour and Minute of the Day at *London*, as was required.

	Hours		Hours.	
As	3921.5	:	48	:: 8765.2 : 107.18
				ho. min. }
			Remains 11	: 10.8 } Requir'd
				hours. }
				or $11\frac{18}{100}$ }

And so for any quantity of time less than one Revolution.

But if the time required to find the Hour and Minute at *London*, (when at Sea or Land,) be some Weeks or Months, *after* it was first set going; or (if cleansed) after it was last *Cleansed*; then you must add 10000.0 to what the two *Indexes* do shew, for each *Revolution*, that it has made, since it was so set going, or last cleansed. And if you would at any time know how many such *Revolutions* must be added, to what is shewn by the Horologe; you must first note the *Day* of the Month, that you are then seeking the Hour for; as suppose *Novem. 12. 1715.* After which note the *Day* last entred, when cleansed; as suppose *Sept. 5. 1715.* Then compute the number of Days between *Sept. 5.* and *Novem. 12.* which is 25, 31, and 12, In all 68. After which observe the *Time* last spent in one Revolution; which you see was 5 Days, 4 Hours, and $\frac{1}{8}$ (for which set down $\frac{1}{10}$ for 'twill make no sensible Difference) and having thus done, say,

	Days	Hours		Hours	Revo.		Days
As	5	4 $\frac{1}{8}$	equal to	124.1	:	1	:: 68 : 13. whole Revolutions;

For which, as before directed, you must add 13 times 10000.0 to the Numbers pointed to, by the two Indexes: Which if we suppose 3752.2 making in all 133752.2 will be the Number required, for the 12th of *November*, if the Indexes should both point as before proposed. And having now found this Number, you are prepared to find the Hour and Minute at *London*, in manner following (*viz.*) suppose you are in or near 44 Degrees of Latitude, in the Autumn Season. Then seek the Latitude 44 Deg. and the Season *Autumn*, in the Book that is peculiar to your Movement, and having found both, take the Numbers Answering thereunto, which (in this Example) is 29274.2 and 240 Hours; and then say,

	Hours		Hours
As	29274.2	:	240 :: 133752.2 : 1096.84

Which being devided by 12, leaves remaining 4.84 ho. or 4 ho. 50.4 min. required, for the Hour and Minute at *London*, by which the Longitude will be found, as before directed.

And seeing it was near the Winter Season: If you should add the Autumn and Winter Numbers, and Hours respectively together; and make them the two first Numbers in your Proportion; it would *Æquate* the Seasons more exactly: I mean, if the said Numbers had been really entred from Practice; but these are only supposed. And by this method you may *Æquate* the Numbers for finding the true Hour and Minute at any Season; or for any other Latitudes whatsoever.

How to find the Hour and Minute, at the place where you are, when abroad, either at Sea or Land.

For Solving this, there are Two Ways, the First and Infallible is, when the Sun is either Rising or Setting; at which times, if the Air be clear, observe what Numbers are pointed to by the two *Indexes*, (but observe the swiftest first, because it will soon be removed) just as the Center of the Sun is in the Horizon; and enter both the Numbers down, as before directed, with the Day of the Month, and the Latitude you are then in: Which having done, by the following Cannon, say,

As P_x

Is to t. c. of the Sun's Decl.

So is t. Lat.

To s. c. of an Angle; which, if converted into Time (by 15) Gives the Hour and Minute of the Sun's Rising from Midnight; and if Subtracted from 12 ho. gives the Setting. Which shews the true Hour and Minute at the Place where you then are, either at Sea or Land.

The Second way for Solving the said Question, is by taking the Sun's Altitude, when about 5 or 10 degrees above the Horizon; and at the same time let another Person see what Number the Swiftest Index points to, and after that the Slowest: Then put them both down as before directed, with the Day of the Month, and the Latitude of the place where you are; then have you the Sun's *Altitude*, the said *Latitude*, and the Sun's *Declination*, to find the Hour and Minute of the Day at the place required.

Now having by the said Observations and Cannons found the Hour and Minute of the Day for the Place where you are, when abroad; and by the Numbers pointed to, by the two *Indexes*, also found the Hour and Minute that it is at *London*, (if the Movement was there set going;) If you take the Lesser of these from the Greater, and bring the Remainder into Minutes, it will give the difference between the two places in Time; by which you may find their difference of Longitude either *East* or *West* to a Minute, by the following proportion, (*viz.*)

Suppose the Difference of Time was 37 Minutes, then say,

$$\begin{array}{ccccccc}
 \{ \text{Minutes of } \} & \{ \text{Minutes of } \} & \{ \text{Minutes of } \} & \{ \text{Minutes of } \} \\
 \text{As } \{ \text{Time } \} & : \{ \text{Longitude } \} & :: \{ \text{Time } \} & : \{ \text{Longitude } \} \\
 \{ 1440. \} & \{ 21600 \} & \{ 37 \} & \{ 555.5 \} \\
 & \text{or 24} & & \text{Deg. Min.} \\
 & \text{hours} & & \\
 & & & \text{or 9 : 15.5} \\
 & & & \text{Required.}
 \end{array}$$

Which 9 deg. 15.5. min. being added to, or taken from the Longitude of the Place, where the said Movement was first set in Motion, gives the Longitude you are come into, as effectually as the Latitude is obtained by Observation, as was at first proposed. So that it is evident, that if any defect should be found in practice, it must proceed from the Vicissitude of the Air; and not from any Deficiency in this Discovery, or from the Cannons relating thereunto.

And that we may Answer all *Objections*, which may arise, on Account of the said Vicissitude of the Air, Motion of the Ship, Foulness of the Movement, &c. as well as render this Discovery more intelligible, we will descend to speak thereof more particularly: And,

In the first place, let Five or Six (or what Number you please) of these Movements be made; and after they have gone for a Month or two, (or if designed for the Sea, 'till within two or three Weeks of your Sailing) then take them asunder, and set them going by the *Meridian* Line, as before directed. And having entred down in the Book, herein after mentioned, the Day of the Month, when set going; how long it is making one Revolution; and what Number is produced, in just Ten Days Motion, (all which must be done, before you Sail in every Voyage;) Let them be carry'd to Sea, into any Latitude, (suppose 50 Deg. or *Newfoundland*) and at their Return bring them again to the said *Meridian* Line, and thereby (with the help of the Number of Days from their being first set in Motion) find the two first *Proportional Numbers*, as before directed: Which done, enter them in a Book, peculiar to each Movement, (the Form whereof followeth) together with the Latitude, and Season of the Year in which the Voyage was made. Then will they all be fitted for a Voyage to *Newfoundland*, and Parts adjacent, for any such Season. And so for any other, when Tried and Entred, as now directed.

After which, send them all in a Ship bound to the Latitude of about 37 Deg. (suppose *Virginia*); and being returned, let the Numbers be found and entred, as

aforesaid, with the Latitude and Season; then will any one, or all of them, be fitted for a Voyage, either to *Newfoundland* or *Virginia*, or any of those Parts, for any such like Season; and so for any other, when Tried and Entred as aforesaid.

This being done, send them also in a Ship, bound to the Latitude of 13 Deg. (suppose *Barbados* or *Jamaica*;) and being returned, find the Numbers, and enter them with the Latitude and Season: So will they all be fitted for these several Latitudes, (*viz.*) 50 Deg. 37 Deg. and 13 Deg. at such a Season: And by a like Process to the *Æquinocial*, and towards the *North Pole*.

Upon which several Tryals, you will have so many Movements fitted for all the *North Hemisphere*, both for Hot, Cold and Temperate Seasons. And, by only Counterchanging the Latitudes, for the whole Globe; or at least for the Marine Parts thereof.

And to what is proposed to be Entred, in respect of Latitude and Season, you may also add the Time of its being first set in Motion after Cleansing; which Cleansing must be done only Annually, and as near as may be on the same Day of the Month; which will much facilitate the Inequality arising from its Foulness: By which all Objections will be answer'd, and the Discovery render'd compleat and perfect. Especially if a true *Meridian Line* be made in every considerable Port in this Kingdom, and elsewhere; for Examining the Movement at your Arrival abroad, and setting it in *Motion*, before your Departure. But remember, that where you set it going, from thence your *Departure* must be reckon'd.

And by this Method all the said Vicissitudes that may arise in respect of Latitude or Seasons, Motion of the Ship (which will be near the same in one Voyage, as in another) and Cleanness or Foulness of the Work, will also plainly appear upon Record, (by comparing one Number in the said Book with another) and will be Accounted for, and *Æquated*, as occasion requires: Namely, by only taking such Numbers from the said Book, to work your Proportion by, as are entred for the Latitude and Season, that you are again Arrived to, or Sailing under: As suppose into 50 Deg. in the Summer Season, then take the Number and Hours answering thereunto; and so, if in Winter, those for the Winter. By which proportionable Numbers, the said Contingencies will be truly and effectually Allowed, and the Difference in Motions justly *Æquated*, as was before asserted.

Note, That in this Movement, the Time is Regulated or Conformed to the Motion: Whereas in former Movements, the Motion is Regulated or

Conformed to the Time. And this ought to be well understood, before you can rightly comprehend what is herein contained.

Now that this Movement, or Discovery, does not only differ from, but also far exceed, any that has hitherto been proposed; is evident in this (*viz.*) That if any former Movement whatsoever be Timed here, as usual, **Before** it is carried to Sea, (tho' never so exactly) yet the change of Climate, &c. will so affect it, as to make it go False: And if you fit it there, then 'twill be also False when brought hither: But if by this Movement, and the foregoing Rules, the said Numbers be found **After** it is returned from such a Voyage; all the said Defects or Contingencies will be *Æquated* by the said Numbers only. So that it will be the same in use, as if no such Accidents did attend it at all. And this Assertion must, and will be granted, by every one that does fully comprehend the Nature and Use thereof. For altho' this *Horologe* has no certain *Hours* or *Time* pre-assigned to it; yet it hath nevertheless, greater or lesser Numbers *Æquated* to the Variableness of Motion or Time, (*viz.*) If fitted to be used here in *England*, then it hath (or may have) one sort of Numbers (or Time) for finding the Hour and Minute, in the Summer Season; another for Winter; and a third for the Spring and Fall; one when it lyeth undisturbed in one and the same place; another (if you please) when carry'd always in a Rattling Coach. And on the other hand, if you carry it to Sea, then it has one for the *Torrid*, one for the *Frigid*, and a third for the *Temperate Zones*, &c. And yet still but one and the same *Horologe*, and but one and the same *Indexes*. So that all manner of Natural Vicissitudes or Contingencies whatsoever, may be Regulated or *Æquated* in and by the said Numbers; and consequently, the true Hour and Minute of the Day, where it is first set going, may be hereby as effectually found, as if no such Impediments did at any time or place attend it, as aforesaid.

And as for what may be suggested in relation to the Inequality between the Spring and Fusie, if any should be, it will be compensated, in that there are five or six Windings up of them, to one Revolution in the *Horologe*. So that there is nothing wanting to compleat this Discovery, but to put it into Practice: And seeing the Charge and Trouble is very Inconsiderable; and the Legislature by *Act of Parliament*, hath been graciously pleased to Incourage hereunto, it is thought expedient to make this publick Proposal, *viz.*

That if any one Person is willing to Engage in this so Noble an Enterprize, he will be readily received as a Partner, on such Conditions as may be Reasonably Concerted therein.

You may hear of the Author at Mr. Burleigh's in Amen-Corner,
London.

MEMORANDUM.

1. All Departures must be reckon'd from *London*, if the Movement was there Set going.

2. That wheresoever they are Set going, there must be a large and exact Meridian Line prepared by a Perpendicular, set up in an Horizontal Circle, as is Mathematically taught, &c.

3. That an Equal time be spent in Winding it up, as near as possible; for which no time need be allowed, if Equated for 10 or 12 Months; because the time required will be the same for the same time, and greater for a greater, &c.

4. That the said Movement be carefully kept in one and the same Temperature of Air, as near as possible.

5. That after this Movement is fully proved, if it should not (by reason of any sudden change of Air) agree to a Minute, yet it ought not in this respect to be altered or much regarded, because the moistness of one Month, may compensate the Dryness of another: So that by such time as you are arrived to the end of your Voyage, the Movement may even Rectifie or Correct it self.

6. It is to be noted, That if the Sun comes too soon to the Meridian, it will also come too soon to the Horizon; and therefore, Watch too fast, or too slow, must be Added or Subtracted, as aforesaid.

7. That it be always wound up, as near as may be, at one certain time of the Day, (*viz.*) if at Sea, about Twylight, or Break of Day, as may be thought expedient. And if it should happen to be omitted, so long as to stand still; you must not set it going again, but by taking it assunder, and putting the two Indexes together, and bringing the Movement to the Meridian Line, as at first directed. Therefore when at Sea be careful of such Omissions.

8. It is to be noted, That altho' every Minute of Time, does Answer to about 15 or 16 Miles *English*, under the *Equinoctial*; yet it is not above 9 or 10 Miles in our Latitude to one such Minute. And altho' it be so many under

the Equinoctial, yet *there* the Sun's Altitude will be taken very exact, because 90 degrees of Altitude, is for 6 hours of Time; whereas in our Latitude, it is but 62 degrees, for 8 hours of Time; both which Advantages will Facilitate the Practice. I shall at present conclude with acquainting,

That if a common Minute Watch (by Filing or Loading the Ballance, or otherwise) be made to go, or does already go, any unknown quantity of Time, either too Fast, or too Slow; by this Device, we may find the Hour and Minute of the Day by such a Watch, more exactly than can be done by the best timed common Watch whatever. And this even without touching it, other than by Winding up the Spring as usual.

There are other Improvements that may be made of this Discovery, but I reserve them to some other Opportunity.

The FORM of the Book to be kept for each Movement, wherein must be entred the Numbers, &c. (which are to be found at the Returning of the Ship) for finding the Hour and Minute at London.

The Movement, Number 1. was first Set Going <i>Sept. 4. 1714.</i> and Then made one Revolution, in 5 days, 4 hours, and a quarter.							
Degrees of Latitude.	Numbers to be found and used in finding the Hour at <i>London.</i>	Hours spent in producing the said Numbers.	Difference in Numbers.	Seasons in which the Ship was Sailing.	Time when the Movement was set in Motion & Try'd; at & after first Making or Cleaning.	Numbers, shewing how the Motion is retarded by Foulness, on a Tryal of 10 Days at a time.	
I.	II.	III.	IV.	V.	VI.	VII.	Days.
50	29276.2	240		<i>Autumn</i>	<i>Sept. 16. 1714</i>	29276.2	10
52	37962.4	432		<i>Winter</i>	<i>Dec. 4. 1714</i>	29272.5	10
44	37942.	430		<i>Spring</i>	<i>Ma. 20. 1715</i>	29264.	10
48	37912.2	431		<i>Summer</i>	<i>June 1. 1715</i>	29237.4	10
Cleansed <i>Sept. 5. 1715.</i> Then made one Revol. in 5 days, 4 ho. and $\frac{1}{8}$.							
50	29274.2	240		<i>Autumn</i>	<i>Sept. 5. 1715</i>	29274.2	10
52	37957.9	432		<i>Winter</i>	<i>Dec. 17. 1715</i>	29268.	10
44	37942.	430		<i>Spring</i>	<i>Ma. 16. 1715</i>	29264.	10
48	37907.5	431		<i>Summer</i>	<i>June 8. 1716</i>	29232.7	10

Note, That the Numbers to be used after *Cleaning*, (*viz.* after *Sept. 5. 1715.*) are found by Subtracting or Adding the Difference between the two Numbers in the Seventh Column, (*viz.* that of *Sept. 5.*) and that of *Sept. 16.* (which is only 2.) out of the former Numbers to be used for the same Latitude; on which there will remain 29274.2 for the *New Number* to be used in Lat. 50. Deg. in *Autumn*, instead of the former, until the next *Cleaning*, (which must be only *Annually*) and so of the rest, from one *Cleaning* to another. And this method must be used for the respective Latitudes, according as Observations shall be made therein, and Numbers found to answer thereunto.

Note also, That this Experiment may be sufficiently Tryed and Proved, by only two or three Voyages, to the South-West parts of *Ireland*, or to *Newfoundland* at farthest.

But notwithstanding what has been said, It is chiefly to be noted, That altho' I have proposed (for Example sake) that the General Numbers should be found, by Sailing to *Newfoundland*, *Virginia*, *Barbados*, &c. and returning again to a *Meridian Line* at *London*: Yet if the *Honourable Persons* before-mentioned in this Title-Page, to whom the whole is most humbly submitted, will please to appoint a Ship for this purpose, the said Numbers may be much better found in manner following; (that is to say) By ordering that the said Ship shall first make a true *Meridian Line*, at or near the Lat. of 60 Degrees; and thereby set the Movements going, in manner aforesaid: After which let them Sail about 40 or 50 Leagues distant from the Shore, and there Cruise *Northward* and *Southward* for about a Month's time; after which let them return again to the same *Line*, and find the Numbers, and also enter them, as before directed. And having so done, let them make another *Meridian Line*, about 15 Deg. more Southward, and there do in like manner: And so at the like Distance to the *Equinoctial*. After which, let them begin again, and repeat the same Process for three or four times: By which means the Numbers will be found, and the different Motion of the Movement discover'd, for all Latitudes and Seasons, as near as needful.

And being returned home, by these Numbers we may find the true *Difference* of Motion, in and for the several Latitudes and Seasons. Which *Difference* must be entred in the 4th Column of the respective Books, reserved for that purpose. By the help of which said *Differences*, you may make as many Movements as may be needful, at any time; as also Books and Numbers peculiar to each Movement. The *Rules* for which will be imparted to any such Watchmaker, as may Reasonably desire the same.

FINIS.

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DISCOVERY FOR FINDING THE LONGITUDE ***

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